

Block and Cumulative Prefix Power Profiles: Exact Constants and the Truncation Tail

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Abstract

All-scale dyadic block bounds and cumulative prefix bounds transfer in both directions with explicit sigma-dependent constants. If small blocks are unavailable, the leftover tail is a real loss and cannot be suppressed by notation. The classification is `DETERMINISTIC_REDUCTION_L1` and the verdict is `PROVED_BIDIRECTIONAL_POWER_PROFILE_TRANSFER`. No program-positive L2 claim or endpoint exponent credit is made.

1 Target contract and source boundary

The target has six simultaneous axes: actual fixed- h_0 packet, source-locked named physical atom, every deterministic prefix, every deterministic scale, a fixed- X power at that atom, and actual active support. The value $h_0 = 2$ is source-backed data only. Repository hashes are integrity locks, not theorem evidence. TPC-193's declared seven-source corpus remains STOP-SCOPED [2].

2 Abstract sequence theorem

Let $0 < \sigma < 1$, $q > 0$, and $A(T) = \sum_{1 \leq n \leq T} a_n$. Write $\Delta(N) = A(2N) - A(N)$, with real endpoints interpreted by integer cutoffs.

Theorem 1 (Power-profile transfer). *If, uniformly for all $N > 0$,*

$$|\Delta(N)| \leq \frac{C}{q} N^{1-\sigma},$$

then

$$|A(T)| \leq \frac{C}{q(2^{1-\sigma} - 1)} T^{1-\sigma}.$$

Conversely, if $|A(T)| \leq (C/q)T^{1-\sigma}$ for all T , then

$$|\Delta(N)| \leq \frac{C}{q}(2^{1-\sigma} + 1)N^{1-\sigma}.$$

Proof. Decompose $(0, T]$ into $(T/2^j, T/2^{j-1}]$, $j \geq 1$, and sum the geometric series $\sum_{j \geq 1} 2^{-j(1-\sigma)} = (2^{1-\sigma} - 1)^{-1}$. The reverse direction is the triangle inequality applied to $\Delta(N) = A(2N) - A(N)$. Under the stated real-endpoint step-function convention the telescoping is exact. An input stated only for integer N requires a separate rounding ledger; it is not silently charged as one endpoint coefficient. $\square$

3 Truncated range ledger

Suppose the block hypothesis starts only at $N \geq M$, and $|a_n| \leq B$. The dyadic decomposition stops when its lower endpoint would fall below M . If R is the remaining endpoint, then $R < 2M$, so the raw tail is strictly less than $2BM$. After the direct normalization q/T , the safe additional charge is

$$\frac{2qBM}{T}.$$

It is small only after a separate range relation makes it small. Thus the full theorem is not a license to identify a terminal block with a cumulative prefix, and the truncated theorem does not yield a fixed X -power for free. This deterministic transfer is distinct from the exceptional-shadow statement of TPC-159 [1].

4 Route consequence

TPC-194 provides the per-packet sequence but not the production schedule or an all-scale block estimate on it. The theorem above is therefore a ready deterministic edge, not endpoint credit.

5 Loss, level and scope ledger

The smallest missing literal theorem is

ALL_SCALE_BLOCK_POWER_BOUND_ON_THE_LITERAL_PHYSICAL_SEQUENCE.

The next route is APPLY_ONLY_AFTER_PRODUCTION_CROSSWALK. No new method cell is stopped in this paper.

The named-atom exponent credit is 0, while the endpoint requires a strict budget greater than $1/400$; the ledger is unpaid. Block sums are not cumulative prefixes, symbolic packet atoms are not named production atoms, phase L^2 and almost-everywhere statements are not pointwise evaluation, and scoped method failures are not theorem or architecture failures.

6 Machine certificate

The adjacent canonical payload freezes the formula or finite witness, exact repository-source hashes, any explicitly non-hashed external locator record, six target axes, scoped stop, route state and claim firewall. Its recursive exact schema has closed objects, exact array positions and constant leaves. The checker recomputes the finite certificate and executes ten adversarial mutations.

References

- [1] Liang Wang. Dyadic shadow prefix lifting, 2026.
- [2] Liang Wang. Literal fixed-atom candidate mechanism gate, 2026.